

Teaching Statement - Lucas Gagnon

I teach by giving students big ideas, small pieces of expertise, and challenging problems that build to mastery. I have learned that intentionally fostering a supportive classroom community catalyzes deep and meaningful learning. I believe that all students can learn, and effective teaching and mentoring helps students become better learners.

This statement will expand on my teaching philosophy and outline my goals, drawing from my experience at **the University of Southern California (USC)**, **York University (YorkU)** and the **University of Colorado Boulder (CU Boulder)**. This includes:

- teaching applied combinatorics at USC, an upper-division course for math and related majors with about 25 students per section (Spring and Fall 2026),
- teaching theoretical linear algebra (1 section with 70 students at YorkU) and introduction to proofs (1 section with 30 students at YorkU), which are both lower-division courses for math and related majors, and
- teaching precalculus (2 sections with 30 students at USC), differential calculus (1 section with 30 students at CU, 1 section with 90 students at YorkU), integral calculus (2 sections with 30 students at CU), and multivariable calculus (1 section with 10 students at CU), which primarily serve quantitatively inclined non-majors.

I have also mentored advanced undergraduate research projects, including:

- an MSRP at USC on combinatorial polynomials, summer projects at YorkU on Hopf algebras and machine learning, and a REU at CU Boulder in group character theory.

Going forward, I plan to grow my experience teaching major-focused courses and cultivating undergraduate research opportunities.

Active lessons: I believe that students build mathematical knowledge by solving problems and explaining ideas. This leads me to use active learning strategies, and to structure lessons as a cycle of engagement, action, and construction. I will illustrate my current approach using a lesson that I taught at USC about the partition type formula for Stirling numbers $S(n, k)$ of the second kind.

I begin by engaging students' curiosity and contextualizing our learning goals. This is often an interactive lecture or group discussion centered around some version of the question "what do we already know, and what are we unsure about?" While the primary goal is to help students engage with new information, it is also an opportunity to informally assess whether the class is prepared for the new topic. In my lesson at USC, I recalled a previous formula for $S(n, k)$, the "multiset formula," and introduced the partition type formula as a computational black box. I engaged students with the question: which formula was better?

After the opening, I transition to small-group discussion, where students practice reasoning and give one another feedback. This is often an informal task like solving a problem from

the board or doing a ‘think–pair–share,’ though I sometimes use tactile exercises and computer interactives. While students work I conduct group check-ins and conduct low-stakes assessments, asking questions like “how do you know this?” and “why?” In my lesson at USC, I asked student groups to first compute $S(6, 4)$ with both formulas, and then to discuss which they preferred. During group work I helped several students assimilate the content of one or both formulas, while noticing that other students had used established principles of counting to interpret the partition type formula.

After a planned interval—usually 4–8 minutes—I bring the class together with group discussion summarizing the activity. This smooths the transition into lecture and helps students practice informal mathematical communication. In my lesson at USC, we identified ease of computation, interpretability, and simplicity as key criterion for a good formula, coming to the consensus that the partition type formula was superior. To support their argument, I asked several students to articulate their combinatorial interpretation of the partition type formula, drawing on what I observed in my earlier assessments. Other students demonstrated critical thinking by asking for an exact enumeration of terms in both formulas.

Finally, I pivot to a summary lecture, where I provide expert insight and model high-level mathematical communication such as writing in complete sentences, signposting, and emphasizing key details (“what,” “how,” and “why”). In my lesson at USC, I expanded on my students’ insight with a formal proof of the partition type formula, and then engaged with the lingering number-of-terms question. This led to a discussion of counting partitions and, without proof, a statement of the Hardy–Ramanujan asymptotic approximation.

I close by connecting the lesson back to the overall objectives in the course, in an informal answer to “what did we learn, and why?” In my lesson at USC, I articulated the importance of mathematics judgement, framing it through current active debates in algebraic combinatorics about the meaning of combinatorial interpretation. I then previewed later concepts in the course that would challenge the class consensus about “good” formulas, including the Inclusion/Exclusion principle and generating series.

Assessing student learning: I use multiple modes of assessment to measure student learning. Because of ready access to generative AI, I combine in-class work with presentation assignments, revision portfolios, and short oral or written explanations that make students’ reasoning visible. For example, in addition to exams I use weekly presentation assignments in my combinatorics class, and have previously used revision-based portfolio assignments and frequent low-stakes quizzes. With each method of assessment, I notice specific improvements over the course of the semester, such as increased confidence in mathematical discussions, deeper metacognition, and improved fluency.

When making an exam or assignment, I write questions to target specific learning objectives and create rubrics to assess student understanding. These practices reduce bias and improve accuracy. I implemented this approach at scale as assistant course coordinator at CU Boulder, where I worked with exam writing committees to create more targeted questions; this experience culminated in the creation of an exam writing guide for future semesters.

Classroom community: I believe it is important to create a welcoming and inclusive space where all students can practice mathematics. Research shows that this improves student belonging and builds long-term peer networks, and I find that it also leads to more collaboration, buy-in, and accountability in my classroom. Implementing concrete strategies accomplish this, such as introducing roles in group discussion, was a key part of my role in the Student Academic Success Center (SASC) at CU Boulder and I continue to use this practice to teach more equitably and effectively.

My approach to building community between students is cumulative. In our first meeting, I ask students to brainstorm norms and behaviors that would make them feel safe taking risks. In subsequent weeks, I encourage students to seek out new working partners, using the first-day norms as a guide for these interactions. After I notice students proactively collaborating, I introduce higher-stakes exercises such as classroom presentations or structured peer review.

Mentorship: Beyond classroom teaching, I have experience supervising undergraduate researchers and graduate teaching assistants. In these roles, I shared the experience of being a professional mathematician by scaffolding research projects and teaching tasks based on my own work. These have been some of my most rewarding teaching experiences: by supporting students as emerging mathematical contributors, I helped them to explore their interest in mathematics and to develop sophisticated ideas and skills.

An additional goal of my mentorship and supervisory work is to produce concrete evidence of student's expertise. While it is subsidiary to learning, accomplishing this effectively has been an area of growth over time. I recently published a paper with undergraduate researchers at YorkU, and am preparing a manuscript with another group based on a summer MSRP at USC. In a pedagogical direction, I helped two undergraduate teaching assistants at CU Boulder develop their own active learning worksheets, which were contributed to a curriculum repository.

Growth: Many of the practices described in this statement were shaped by prior challenges, and I will continue to adapt and improve as I gain experience. Early in my teaching, I found that unstructured group discussion brought out the same confident voices. I responded by introducing individual preparation time, replacing volunteer-only participation, and following-through with whole-class synthesis. These structures broadened participation while preserving the independence I wanted students to develop.

I take concrete actions to improve my teaching, including completed a teaching intensive at YorkU and a teaching certificate program at CU Boulder. I also watch my colleagues teach and solicit student feedback to better understand the practice of teaching and find new ideas. Finally, I regularly develop new lessons, materials, and assessments to connect my students with this ongoing growth.

Conclusion: Although I continue to learn, my experience makes me comfortable using active learning in a range of environments. I am committed to equitable teaching practices. I am excited to explore new challenges and achieve excellence in all aspects of teaching.